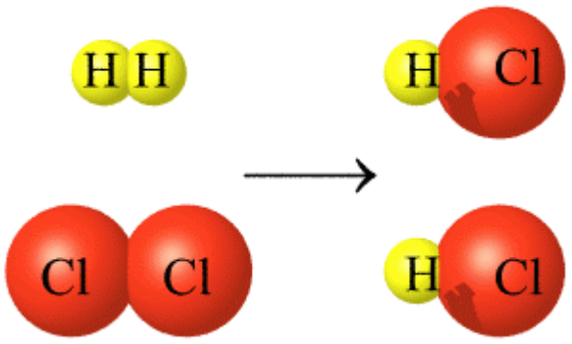
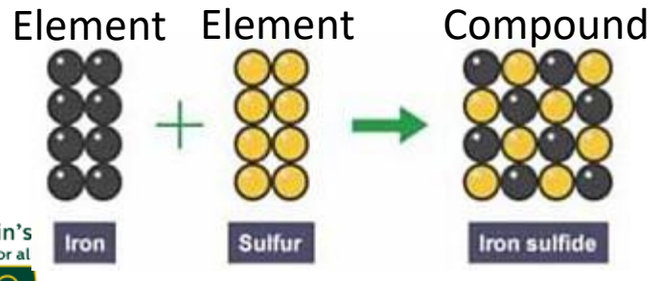
Knowledge Organiser
Year 8: Chemical Reactions - Compounds

Chemical vs Physical Changes
- When a chemical reaction occurs, atoms are neither created or destroyed
- Instead the atoms rearrange themselves, forming new bonds between atoms
- Frying an egg and neutralisation are chemical changes
- A physical change occurs when particles are rearranged but no new substances are made
- Changes of state and dissolving are physical changes

Particle diagram which shows the formation of **new bonds between atoms** during a chemical change

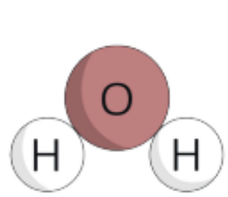


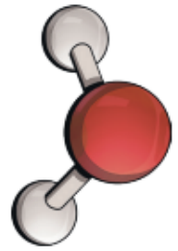
What is a compound?
- A chemical compound is formed when two or more elements are **chemically combined**
- The chemical compound will have different properties to the elements it made from
- Sodium chloride (salt) is a white **crystalline compound** formed from sodium, a silvery reactive metal and chlorine, a yellow green gas



How can we represent a chemical compound
- The atoms in a chemical compound are chemically bonded
- We can show this in lots of different ways
- A formula tells you how many atoms of each element there are
- CO_2 is the formula for carbon dioxide
- We can also use particle diagrams like the ones below

Bonds are shown with the atoms touching

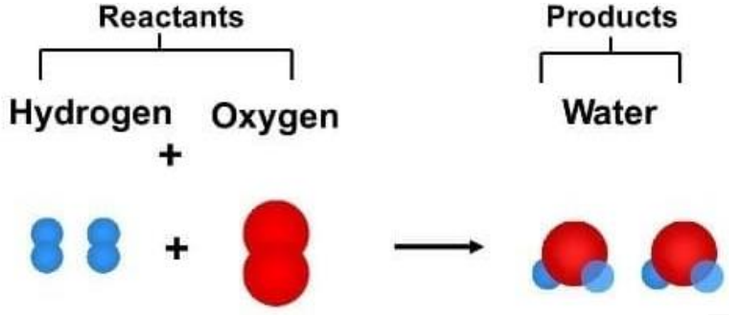




Bonds shown as sticks between the atoms

How do we write chemical equations?
- A chemical equation shows how reactants are turned into products
- The reactants are the substances which react together
- Reactants are shown on the left of the arrow
- The products are the substances which are made
- Products are shown on the right of the arrow

A labelled diagram showing the reactants and products in the reaction between hydrogen and oxygen



How do we name chemical compounds?
When elements react together, the compound has a name which includes all the individual elements

Rule 1: The metal goes first in the name
e.g. Sodium + Chlorine → Sodium (metal) chloride (non-metal)

Rule 2: If there are just two elements the name of the compound ends in –ide, e.g. sodium chlorIDE

Rule 3: If there is a non-metal and oxygen present then the compound ends in –ate e.g. sodium chlorATE

Formulae	Name of compound
$MgCl_2$	Magnesium chloride
$NaNO_3$	Sodium nitrate
Fe_2O_3	Iron oxide
$MgCO_3$	Magnesium carbonate
BeS	Beryllium sulphide
$CuSO_4$	Copper sulphate

What is a balanced chemical equation?
- A balanced equation is one where mass is conserved
- The number of atoms of each element must be the same in the reactants and products

An equation showing the **unbalanced equation** for the reaction between iron oxide and carbon

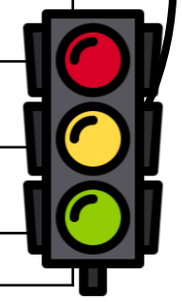
$$Fe_2O_3 + C \rightarrow Fe + CO_2$$

Reactant side	Product side
Fe - 2	Fe - 1
O - 3	O - 2
C - 1	C - 1

This is **unbalanced** because from reactants to products, an Iron and oxygen atom are lost.

RAG Yourself!

Learning Outcomes;	RAG
Describe the difference between chemical and physical changes	
Describe the difference between compounds and mixtures	
Explain why the properties of elements change when they react together	
Describe what the different parts of a chemical equation mean	
Write chemical word equations when given the reactants	
Balance chemical symbol equations	



Knowledge Organiser
Year 8: Chemical Reactions - Compounds

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Particle diagram which shows the formation of **new bonds between atoms** during a chemical change

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Bonds are shown with the atoms touching

Bonds shown as sticks between the atoms

How do we write chemical equations?

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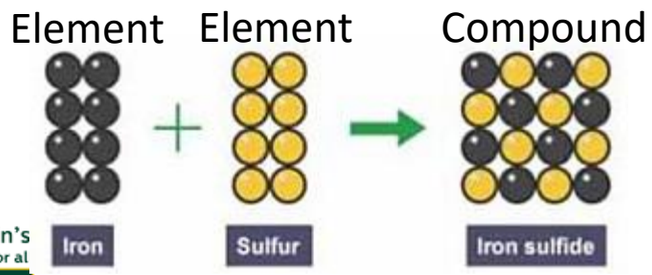
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